

RENANCE

AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 306 — Past Questions & Materials

by Renance

Course: CSC 306

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — CSC 306

Level 300 • First Semester • PastQuestion

AMBROSE ALLI UNIVERSITY, EKPOMA

FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION FOR 2019/2020 ACADEMIC SESSION

Course Code: CSC 306 - Systems Analysis and Design

Instruction: Answer all questions in Section A and 3 questions in Section B.

Time allowed: | hour 45 minutes

SECTION A ~ COMPULSORY (30 Marks)

1. System Study involves (a) study of an existing system (b) documenting the existing system (c) identifying current deficiencies and establishing new goals (d) All of the above (e) None of the above
2. The primary tool used in structured design is (a) structure chart (b) data-flow diagram (c) program flowchart (d) module (e) None of the above
3. In a __ one module of the new information system is activated at a time.
(a) System Development Life Cycle (b) CASE tool (c) Phased Conversion (d) Success factors (e) None of the above.
4. In Prototyping is used (a) BASIC (b) COBOL (c) 4GLs are used (d) system is documented (e) None of the above
- 5: Which of the following is not a factor in the failure of the systems developments projects? (a) size of the company (b) inadequate user involvement (c) failure of systems integration (d) continuation of a project that should have been cancelled (e) None of the above
6. A problem's will answer the question, "What information will the computer need to know in order to either print or display the output times?" (a) Input (b) Output (c) Processing (d) Purpose (e) None of the above
7. To run the old system and the new system at the same time for a specified period, the system implementation approach used is (a) pilot (b) phased (c) parallel (d) direct (a) None of the above
8. The mistake, committed by interchanging two digits in a numeric held, during data entry, is called (a) transposition error (b) transcription error (c) Beta testing error (d) Alpha testing error (e) None of the above
- 9, Problem analysis is done during __ (a) system design phase (b) systems analysis phase (c) before system test (d) All of the above (e) None of the above
10. Documentation is prepared (a) at every stage (b) at system design (c) at system analysis (d) at system development (e) None of the above
11. System Implementation Phase entails (a) System check outs (b) Pilot

run (c) Parallel runs (d) All of the above (e) None of the above

'allowing is not true of the conversion phase of the development life

18 personnel must work closely together, (b) steps must

(c) documentation should be emphasized (d) the

tem should be considered (e) None of the above

12, — Which of the |

evele? (a). the user and system

be taken to phase out the old system

non machine components of the sys

potential users of the MIS _ (a) select the optimum

tions (a) evaluate alternate equipment configurations (c) describe

of the above (e) None of the above

13. Managers who are

equipment configura

information needs (d) All

nt life cycle, which of the following aspects are

14, In phase | of the system developme

nsactions) (c) controls (d) All of the above

usually analyzed? (a) outputs (b) input trar

(e) None of the above

___ (a) System requirements are established (b)

18. During the maintenance phase ___

Programs are tested (d) All of the above (e) None of

System analysis is carried out (Cc)

the above

SECTION B: Attempt any 3 (15 marks Each)

(2)a. What is an Information system, why is control necessary ina system? (5 marks)

b. List the elements of a system and explain 5 of them. (5 marks)

c. Who are the people involved in System Analysis, discuss five (5)

characteristics of a system. (3 marks)

(3)a. List and explain five various fact finding techniques. (5 marks)

b. In system design, two stages are involved, list and discuss the two stages. (5 marks)

c. Explain the various approaches to system conversion along with their

advantages and disadvantages.

(5 marks)

(4)a. What is a Prototype? With the aid of a diagram, explain the developmental

(5 marks)

process of a prototype.

b. What is system investigation

investigation?

c. Explain the following terms: (i) batch processing (ii) on-line systems

(iii) Knowledge- based system (iv) decision-support system

(v) real-time systems

2 Who are the people involved in system

(5 marks)

(5 marks)

(5)a. What are software analysis and design tools? (5 marks)

b. What do you understand by fourth generation

categories of the fourth-generation tools? (5 marks)

c. Briefly discuss the following; (i) Software user interface (ii) command line interface

(iii) graphical user interface (iv) GUI elements (v) GUI implementation tools (5 marks)

languages? What are the

(6)a. What is a Data Flow Diagram? Differentiate between DFD and Flowchart. (5 marks)

es List and discuss two generic approaches to software designing (5 marks)

c. Discuss the following terms (i) CASE tools (ii) encapsulation (iii) polymorphism

(5 marks)

(iv)objects (v) classes

Good Luck!

AMBROSE ALLI UNIVERSITY, EKPOMA

FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION FOR 2019/2020 ACADEMIC SESSION

CSC 306 - Systems Analysis and Design

tion A and 3 questions in Section B.

Date: 11" March, 2021

Course Code:

Instruction: Answer all questions in Sec

Time allowed: | '2 [rs

SECTION A COMPULSORY (30 Marks)

ystem (b) documenting the existing

(a)study of an existing §

(q)All of the above

Ls System Study inv olves

deficiencies and establishing new goals

system (¢) identifying current

(c) None of the above

wed design is (a) structure chart (b) data-flow

c

2 The primary tool used in structure

module (c) None of the above

diagram (c) program flowchart (d)

one module of the new information system is activated at a time.

3. In a

b) CASE tool (c) Phased Conversion (d) Success factors

(a) System Development Life Cycle (

(e) None of the above.

Vv

4 In Prototyping, is used (a) BASIC (b) COBOL (c) 4GLs are used (d) systems

documented (e) None of the above

failure of the systems development projects?

ent (c) failure of systems integration

lled (e) None of the above

>. Which of the following is not a factor in the

(a) size of the company (b) inadequate user involvement

(d) continuation of a project that should have been canceled

6. A problem's —_ will answer the question, "What information will the computer need to know in order to either print or display the output times?" (a) Input (b) Output (c) Processing

(d) Purpose (c) None of the above

new system at the same time for a specified period, the system

dq: To run the old system and the

(8) pilot (b) phased (c) parallel (d) direct

implementation approach used is _

(e) None of the above

d, during data entry, 's

8. The mistake, committed by interchanging two digits in a numeric helix, is called a testing error (d) Alpha

called a) transposition error (b) transcription error (c) Beta

testing error (c) None of the above

(a) system design phase (b) systems analysis

a. Problem analysis is done during

of the above

phase (c) before system test (d) All of the above (e) None

10. Documentation is prepared (a) at every stage (b) at system design

analysis (d) at system development (c) None of the above

esign (cat system

11. System Implementation Phase entails (a) System check outs (b) Pilot run

(c) Parallel runs (d) All of the above (c) None of the above

12. Which of the following is not true of the conversion phase of the development life cycle?
 the user and systems personnel must work closely together. (b) steps must be taken to phase out the old system (c) documentation should be emphasized (d) the non machine components of the system should be considered (e) None of the above

13. Managers who are potential users of the MIS (a) select the optimum equipment configurations (a) evaluate alternate equipment configurations (c) describe information needs (d) All of the above (c) None of the above

14. In phase 1 of the system development life cycle, which of the following aspects are usually analyzed? (a) outputs (b) input transactions (c) controls (d) All of the above (e) None of the above

(e) established (b) System

(a) System requirements are

(e) of the above

15. During the maintenance phase

tested (d) All of the above (c) Non

analysis is carried out (c) Programs are

SECTION B: Attempt any 3 Questions (15 marks each)

Q1. a. What is an Information system, why is control necessary in a system? Poy (c- List the elements of a system and explain any 5 of them. Pay« B/F SA (5 marks (5 marks

b.

c. Who are the people involved in System Analysis, discuss five (5) characteristics of a system. page 6 [7e 43/24 (5 marks, > marks)

(3) a. List and explain five various fact finding techniques. 3 ! X 32

b. In system design, two stages are involved, list and discuss the two stages. 24a3S (5 marks)

c. Explain the various approaches to system conversion along with their advantages and disadvantages. Pra 43 «4. 45 (5 marks)

y /

(5 marks)

(4) a. What is a Data Flow Diagram? Differentiate between DFD and Flowchart. 7), << (5 marks)

b. List and discuss two generic approaches to software designing

c. Discuss the following terms (i) CASE tools (ii) encapsulation (iii) p

(iv) objects (v) classes

polymorphism

(5 marks)

(5)a. What are software analysis and design tools? Ty (5 marks)

b. What are fourth generation languages and what are the categories of the fourth-generation tools? 61 % Ye (5 marks)

c. Briefly discuss the following: (i) Software user interface (ii) command line interface (iii) graphical user interface (iv) GUI elements

(v) GUI implementation tools 61 G2 ,64%, 65, ¢) (5 marks)

» With the aid of a diagram, explain the developmental process (5 marks)

Who are the people involved in system investigation?(5 marks)

(6)a. What is a Prototype of a prototype.

b. What is system investigation

Explain the following terms: (i) batch processing (ii) on-line systems

(iii) Knowledge-based system (iv) decision-support system (v) real-time systems (5 marks)

9 x AMBROSE ALLI UNIVERSITY, EKPOMA '

Me FACULTY OF PHYSICAL SCIENCES

" DEPARTMENT OF COMPUTER SCIENCE

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FIRST SEMESTER EXAMINATION FOR 2018/2019 ACADEMIC SESSION

a CSC 306: SYSTEMS ANALYSIS AND DESIGN

ne Instruction: Answer QUESTION 1 and any other THREE Questions

F

Time: 2HRS

Date: 11" June 2019

QUESTION 1

'® # What is a system? List and explain the three common types of systems giving three examples each. (5 marks)

@ What do you understand by the term system? (3 marks)

a) What is Systems Analysis and why is it necessary? (3 marks)

"System Control"? Why is control necessary in a

&@ Various players are involved in the development of 2 system, list the major categories of ~" players and explain any five of them. (8 marks) :

ey Who is a

Systems Analyst? Discuss the Characteristics of a Systems Analyst. (6 marks)

QUESTION 2

a) What is System Development Life Cycle (SDLC)? (2 marks)

A ' With the aid of a diagram, explain the various phases of SDLC (6 marks)

φ) List and explain the various fact finding techniques (7 marks)

, / _ QUESTION 3

* a) What do you understand by software

φ analysis and design tools? (2 marks)

a) Discuss in detail 5 different types of software analysis and design tools (5 marks)

c) Define conversion: List the various steps in file conversion; (5 marks)

y d) What is system maintenance and how does it work (3 marks)

QUESTION 4 |

/ a) In your own understanding, discuss Prototyping as an approach to system development (3 PA marks)

b) What are the Steps involved in building

c) What are the

@) Discuss 4 applications

§ a prototype (4 marks)

merits and demerits of Prototyping (4 marks)

Citations that are good Prospects for prototyping (4 marks)

\ i" 4 . QUESTION 5

4 NY What is a flowchart? List the different types of flowcharts. (4 marks)

J With diagrams, discuss 5 basic flowchart

3) .

Q — List and explain the

2) What is object? Diff QUESTION 6

x What is an object? Differentiate between an object and system (8 marks)

b) Define the following terms: object and 'an entity, (3 marks)

(i) Objects and Classes (ii) Inheritance (iii) polymorphism (3 marks)

i o l } Online

Processing (vi) CASE (12 marks) polymorphism (iv) Batch processing (v)

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DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF PHYSICAL SCIENCES

AMBROSE ALLI UNIVERSITY, EKPOMA
 2016/2017 FIRST SEMESTER EXAMINATION.
 ' CSC 306; System Analysis and Design

"INSTRUCTION: ANSWER QUESTIONS ONE AND ANY OTHER THREE TIME: 2 "HRS

, 1) What is 2 system? Explain the concepts; systems analysis and systems design (6marks)

(0) State six attributes of a system (6marks)

(c) Give five reasons why an existing syst m may need revision. (5 marks)

(d) State five characteristics of a goad s¥stem analyst. (5 marks) ,

(ec) Use a diagram only to illustrate'the stages involved in system development life cycle. (marks)

eee ot

_ 2(a) What is 'easibility study? (dmarks) ee secs hae aa

(b) List and explain the different types of feasibility study in systems developmen (7m a
 duadvatcion:

(c) What is an interview? List'and explain two main types of interview, give their advantages an a
 eo (9 marks) ° .

bw oo ——— us

3 (a) What is Compiiter-based information System? G3 marks) oo"

. (b) State the three types and explain them as detailed'as possible (6marks

(c) Why is maintenance of:a system necessary? (3 marks) ; .

O Write short notes on three facts gathering techniques indicating two advantages and the
 disadvantages in each

* technique. (6 marks) : .

24 An engineering firm is tendering for a contract to supply a steel fabrication with target duration
 of

20 weeks. The tasks have been analyzed as follows.

{_ ACTIVITY __IMMEDIATEPREDECESSOR _| DURATION (days)

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J Cc | 10 |

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(a). Draw the network diagram, calculate the expected project duration, identify the critical path using event

analysis method and calculate earliest event time for activity H (10 marks)

(b). State two titles under which network analysis is operated. (4 marks)

(c) What is a sub-system? Use a real life business organization to explain a sub-system. (4 marks)

5 =") Identify four types of tools used in-systems-analysis-and-describe One OT Mei in detail (6marks)

%b) Identify four actors and their roles in software development. (6 marks) a

ao

(c) List the major activities performed in system study phase (6marks)

» 6{a) What is System Conversion? (2marks)

{b) Discuss the different approaches of system change over and give an instance under which each approach is

preferable to other approaches (8marks)

{c) Discuss the different classes of a system and state the relationship between a computer based system and the

natural systems (8 marks) ; ;

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AMBROSE ALLI UNIVERSITY, EKPOMA'

FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION FOR 2017/2018 ACADEMIC SESSION

CSC 306: SYSTEMS ANALYSIS AND DESIGN | Time: 2HRS

Instruction: Answer QUESTION 1 and any other THREE Questions Date: 3" April 2018

of QUESTION 1

a) Discuss your understanding of Systems Analysis and Design? What is the goal of Systems Analysis and Design? (4 marks)

"d) List and explain two types of systems (2 marks)

J c) Differentiate between Systems analysis and System Design (2 marks)

24) List and explain five stakeholders in a System. (5 marks)

2) Discuss 5 duties of a System Analyst. (5 marks)

~ Discuss the three basic concepts of a system (3 marks)

Yg) Why is control necessary in systems? (2 marks)

oh) Why does system change? (2 marks) ~ 7

QUESTION 2

vv <a) Wh-t do you understand by System Development Life.Cyclé (SDLC)? (2 marks} ~

.e

= .. Witi the aid of a diagram, explain the phases of SDLC (10 marks)

c) Discuss the three types of feasibility in SDLC (3 marks)-

/ QUESTION 3

a) Lisi and explain 5 types of fact finding techniques (5 marks)

b) Def ne conversion. What are the four common approaches to system conversion (5 marks)

c) Wh: t is system evaluation? (2 marks)

d) Discuss your understanding of system maintenance and how it works (3 marks)

QUESTION 4

"a) Whet is s Soweheart? Explain the different types of flowcharts. (5 marks)

b) Defae Data Flow Diagram (DFD)? With the aid of diagrams, explain the different .

corm sonents of DFD. (5 marks)

ç) Exp ain the various types of Connectivity and cardinalities (3 marks)

d) Wh: t do you understand by Computer-Aided Software Engineering (CASE) (2 marks)

QUESTION 5

#) Wh. tis Prototyping and what are the objectives of prototyping? (5 marks)

2 Sei ain the process involved in designing a prototype system. (5 marks)

uss 5 advantages and disadvantages each of prototyping. (5 marks)

QUESTION 6

Dis ç j Be j

. Su = following terms: (a) Objects and Classes (b) Method and Message (c) Inheritance

Pecessing (€) Online processing (15 marks)

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srineranan ENT OF COMPUTER SCIENCE,

te ORATE OF SCIENCE AND HUMANITIES,

BROSE ALLI UNIVERSITY, EKPOMA.

20192024 SECOND SEMESTER EXAMINATION.

s€ 306: Systems Analysis and Design DATE: 07-12-2021

INSTRUCTION: Answer Question ONE and Any Other THREE TIME: 2 HRS

J i@v What is a system? Explain the concepts: systems analysis and systems design (4 marks)

(bv State four characteristics of a system (2 marks) ;

ON Discuss with examples. three ty pes of information system (6 marks)

(@) Give four reasons why an existing system may need revision. (4 marks)

(e)y Stale five characteristics of a good system anaiyst. (5 liarks)

(Nv Use a diagram only to illustrate the stages involved in system de

(g)/ Explain two types of system (2 marks)

velopment life cycle. (2 marks)

necessary in system development? (4 marks)

marks)

unique. (6 marks)

V What is a feasibility report? Why is feasibility study n
nt types of feasibility study in systems development. (3
ate the advantages and disadvantages of interview tec
(2 marks)

2 (a)

(b) ¥ Explain three differe

(c)y What is an interview? St

(d) y Discuss four methods of file organization

ystem? (2 marks)

3(a) | What is Computer-based Information S

(b) Give three reasons why maintenance of a system is necessary (3 marks)

(c) Identify four actors and their roles in software development. (6 marks)

(d) List the major activities performed in system study phase (4 marks)

analyst what is your perception of prototyping (3 marks)

4(a) As a system

a prototype system (2 marks)

(b) yy Discuss the objectives of

(c) V Explain the following terms

(i) "Method and message (2 marks) >

(ii) vObject Identity (old) (2 marks)

(iin Classes (2 marks) =

(iv) Object state (2 marks)

(v) Mnheritance (2 marks)

scribe two of them in detail. (5 marks).

ools used in systems analysis and design and de:

ing the network diagram. (6 marks)

5(ai) Identify six types of

(ii) As a system analyst, model the following task analyzed below by draw
er: eee ee

ACTIVITY. ___|___ PRECEDING ACTIVITY | DURATION (weeks)

pe ee (DINGACHIVITY Ceo ooo

at Determine the paths (tough tie network (2 marks)

iv) — Identify the critical path and calculate its duration (2 marks)

6 a alee is simple". Do you agree? Justify (3 marks)

discuss the different approaches of sy ;

: of system change over and give an ins i

4 meta alain "abe and give an instance under which p

-) State five me' 5 Ol Inlere ith

Oe i olan of interacting with a system and explain any two. (5 marks)

ss two software design strategies (2 marks)

arallel approach is

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AMBROSKE ALL NATURAL SCI

2015/2010 Hitt ae UNIVERSITY BPOMA 5

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SC 300: Las GR@ANISATION Super Arey DEANE vs

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i a diagram only to (usteate the slayos lavelody tin nyelers developrighnt (marks) 6s t fey beyans (wa

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